Intro to AI

(This coding assignment has two parts. One part is take-home (15 points) and the other part will be in class (5 points))

In the take-home part of the assignment, we are going to develop a deep neural network from scratch on an open-source tabular dataset, benchmark it against an AutoML system, and deploy the better model as a working inference service.

1. Obtain the UCI Adult Census Income dataset (https://archive.ics.uci.edu/dataset/2/adult) or load it with OpenML (dataset id 1590). Build a reproducible preprocessing pipeline: handle missing values, encode categorical features, scale numeric features, and create a stratified train/validation/test split with a fixed random seed. Report the split sizes, the class balance, and every preprocessing decision you made. (3 points)
2. Implement a deep feed-forward neural network from in PyTorch or Tensorflow — your own architecture, loss, optimizer and training loop, with no pretrained models and no AutoML at this step. Use at least two hidden layers. Report: (3 points)
    a. The architecture table (layers, units, activations) and the total number of trainable parameters.
    b. Training and validation loss/accuracy curves.
    c. Test accuracy, precision, recall, F1 and ROC-AUC, plus the confusion matrix.
3. Tune the network manually. Train at least eight configurations that vary depth and width, learning rate, batch size, and regularization (dropout, weight decay, batch normalization, early stopping). Report a table of all configurations with validation metrics and training time, and state which factor mattered most and why. (3 points)
4. Run an AutoML framework — AutoGluon Tabular, FLAML, or auto-sklearn — on exactly the same split. Report the leaderboard, the selected model and its hyperparameters, and compare it against your hand-built network on accuracy, F1, ROC-AUC, training time, model size on disk, and inference latency. Compare step 2 with the AutoML system. (3 points)
5. Deploy the better of the two models. Export the trained model together with its preprocessing pipeline, write a predict function that accepts raw unprocessed input, and wrap it in a FastAPI application running in Colab or on localhost. Demonstrate five sample predictions through the interface, measure the average end-to-end latency per request, and describe how you would handle a feature the model has never seen at inference time. (3 points)
6. Write a report in a word document by responding to each step. You should report every assumption you made during the project (with reasoning), every tool you have used (with reasoning), and screenshot of your running deployment. Submit the document and code into a zip file.

Note: You can use any AI tool or any available resources for finishing the assignment. It is expected that you understand each steps in details and the report you submit should only contain your views and experimental findings.